Unit 2, Lesson 5: Approximating Irrational Numbers

Benchmarks

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

MA.8.NSO.1.2 Plot, order, and compare rational and irrational numbers represented in various forms.

Learning Targets

- ☐ I can identify whether a number is rational or irrational.
- ☐ I can approximate the values of irrational numbers.
- ☐ I can plot irrational numbers on a number line.

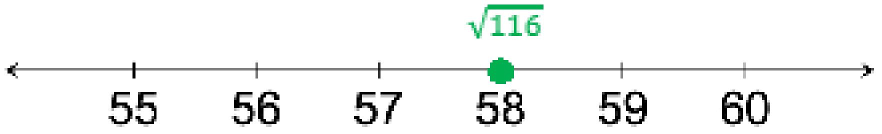
2.5.1 Warm-Up: Rational vs. Irrational

1. Complete the table by filling in values in each box with a digit, 1 through 8, such that the value of the expression matches the given description. Each digit can only be used once.

Expression	Description
$\sqrt{\square}$	This is a rational number.
$\sqrt{\square}$	This is an irrational number.
$\frac{\square}{\square}$	This is a rational number and an integer.
$\frac{\square}{\square}$	This is a rational number with a terminating decimal expansion.
$\frac{\square}{\square}$	This is a rational number with a repeating decimal expansion.

2.5.2 Collaboration: Error Analysis

Robin is learning about square roots. A question and her response are shown in the table. Since this is new for Robin, there are some mistakes in her response.

Question	Robin's Response
Where would $\sqrt{116}$ be located on a number line?	*Taking the square root is the inverse of squaring a number. Squaring a number is the same as raising it to the second power. To find the value of a square root, you divide the radicand by 2. The number line shows the location of* $\sqrt{116}$ *because* $116 \div 2 = 58$. 

1. **Discuss Robin's response with your partner. Then, cross out any incorrect statements in her response.**

2. **Work with your partner to write a correct explanation of how to locate $\sqrt{116}$ on a number line.**

3. **Use the number line to plot and label a point to better approximate the location of $\sqrt{116}$. Be sure to write an appropriate scale using the tick marks.**

2.5.3 Guided Instruction: Rational and Irrational Numbers

The real number system includes both ***rational*** and ***irrational numbers***.

A ***rational number*** is a real number that can be expressed as a ratio of two integers.

Rational numbers can be written as a fraction, $\frac{a}{b}$, where a and b are integers and $b \neq 0$, or as a decimal. In decimal form, rational numbers either terminate or repeat.

An ***irrational number*** is a real number that cannot be expressed as a ratio of two integers.

When expressed as decimals, irrational numbers do not terminate or repeat. Therefore, any decimal expansion is only an approximate value.

1. **The relationship between the circumference and diameter of any circle is represented by pi (π).**

   a. **Discuss with your partner what you remember about the value of π.**

   b. **Complete the statements.**

   The ratio of the circumference of a circle to its diameter is expressed as π. The values 3.14 and $\frac{22}{7}$ are

   - o equivalent to, or the exact value of
   - o used to approximate the value of

   π.

   The exact value of π

   - o can
   - o cannot

   be expressed as a fraction of two integers or a decimal. Therefore, π is

   - o a rational
   - o an irrational

   number.

2. Enter each of the following values into a calculator.

$\sqrt{25}$ $\sqrt{18}$ $\sqrt[3]{27}$ $\sqrt[3]{30}$

a. Circle the values that can be expressed as a ratio of two integers.

b. Discuss with your partner how you would describe the values that are not circled.

c. Complete the statements.

Square roots of perfect squares and cube roots of perfect cubes
- o can
- o cannot

be expressed as a ratio of two integers. Therefore, they are
- o rational.
- o irrational.

Square roots of non-perfect squares and cube roots of non-perfect cubes
- o can
- o cannot

be expressed as a ratio of two integers. Therefore, they are
- o rational.
- o irrational.

2.5.4 Collaboration: Rational or Irrational?

Work with your partner to complete the following.

Several numbers are given:

$$-\frac{4}{5} \qquad \frac{\pi}{2} \qquad 5.124784\ldots \qquad 4.\overline{35} \quad 3.275 \quad -\sqrt{40} \quad \sqrt[3]{-27}$$

1. **Write each expression in the appropriate column of the table based on the definitions of rational and irrational numbers.**

Rational	Irrational

2. **Explain how you determined whether $\frac{\pi}{2}$ is rational or irrational.**

3. **Explain how you determined whether $4.\overline{35}$ is rational or irrational.**

2.5.5 Guided Instruction: Approximating Irrational Numbers and Locating Them on a Number Line

The values of irrational numbers and expressions involving irrational numbers can be approximated.

1. Consider the expression $\sqrt{13}$.

   a. Between which two whole numbers does $\sqrt{13}$ lie on a number line?

   b. Which whole number is $\sqrt{13}$ is closer to?

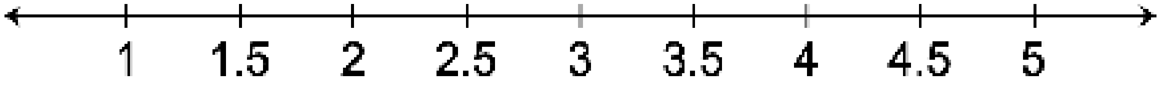
   c. Faith thinks the value of $\sqrt{13}$ will be close to 3.6 or 3.7. Discuss with your partner why Faith may have made these estimates.

   d. Check Faith's estimates to determine if either estimate is close to the value of $\sqrt{13}$.

   e. Based on your work in Part D, complete the statement.

   The approximate value of $\sqrt{13}$, to the nearest tenth, is ____.____.

   f. Plot and label $\sqrt{13}$ on the number line to indicate its approximate location.

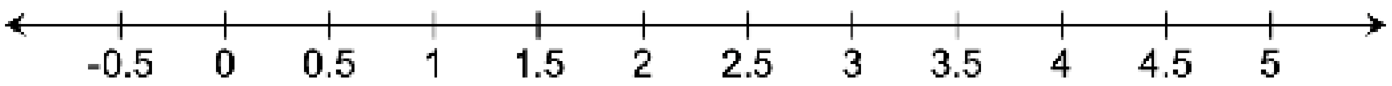


2. Use the approximation of $\sqrt{13}$ from the previous question to complete the following.

a. Approximate the value of each expression.

Expression	Approximate Value
$\sqrt{13}+1$	
$\frac{\sqrt{13}}{2}$	
$4-\sqrt{13}$	

b. Plot and label the approximate location of each expression from part a on the number line.



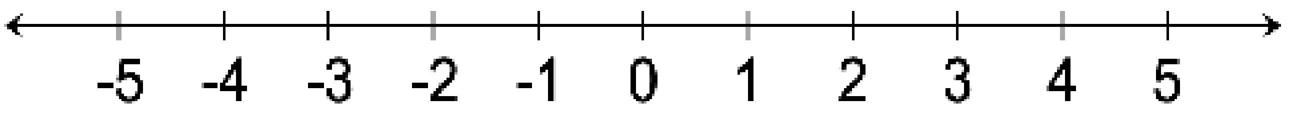
To approximate the value of expressions involving irrational numbers, first approximate the value of the irrational number. Then, use that approximation to perform any operations indicated.

2.5.6 Guided Practice: Approximating Irrational Numbers and Locating Them on a Number Line

1. Approximate the value of each expression.

Expression	Approximate Value
$-\pi$	
$\sqrt{6}$	
$\sqrt[3]{20}$	
$2 \cdot \sqrt{6}$	
$\pi - 3$	

2. Plot and label the approximate location of each expression from the previous question on the number line.



2.5.7 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the lesson that I knew were...	Something I wondered during the lesson was...	Something I learned during lesson was...